Carbon Dioxide Protects the Perinatal Brain From Hypoxic-Ischemic Damage: An Experimental Study in the Immature Rat

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ABSTRACT. *Background and Objective.* **Clinical investigations suggest that premature infants who require mechanical ventilation from respiratory distress syndrome are at increased risk for periventricular leukomalacia if hypocapnia occurs during respiratory management. The question remains as to the contribution of hypocapnia to hypoxic-ischemic brain damage and whether or not hypercapnia is neuroprotective.**

Methods. **Seven-day postnatal rats underwent unilateral common carotid artery ligation followed thereafter by exposure to systemic hypoxia with 8% oxygen (O_2) combined with either 0, 3, 6, or 9% carbon dioxide (CO_2) for 2.5 hours at 37°C. Survivors underwent neuropathologic examination at 30 days of postnatal age, and their brains were categorized as follows: 0 = normal; 1 = mild atrophy; 2 = moderate atrophy; 3 = atrophy with cystic cavitation <3 mm; 4 = cystic cavitation >3 mm of the cerebral hemisphere ipsilateral to the carotid artery ligation. The width of the ipsilateral hemisphere also was determined on a posterior coronal section and compared with that of the contralateral hemisphere to ascertain the severity of cerebral atrophy/cavitation. Data were analyzed by linear models.**

Results. **CO_2 tensions averaged 26, 42, 54, and 71 mm Hg in the 0, 3, 6, and 9% CO_2 exposed animals, respectively, during systemic hypoxia. Blood O_2 tensions during hypoxia were not different among the four groups and averaged 34.7 mm Hg. Neuropathologic results showed that 30/38 (79%) rats exposed to 3% CO_2 showed either no or mild brain damage compared with 13/33 (39%) controls (0% CO_2). Cystic cavitation occurred in only four CO_2 exposed rat pups compared with 14 controls (P = .001). At 6% CO_2 exposure, all of 20 rat pups showed either no damage or mild atrophy compared with controls (P < .001); and at 9% CO_2 exposure, 19/23 (83%) rat pups showed no or mild damage compared with controls (P < .001). The data also showed that the greatest reduction in brain damage occurred in immature rats exposed to 6% CO_2 with slightly less protection at 9% CO_2 (P = .012), the latter comparable with the severity of brain damage sustained by animals inhaling 3% CO_2. Analyses of coronal width ratios at each CO_2 exposure provided results comparable with those of the gross neuropathology scores.**

Conclusions. **The results indicate that in an immature rat model normocapnic cerebral hypoxia-ischemia is associated with less severe brain damage than in hypocapnic hypoxia-ischemia and that mild hypercapnia is more protective than normocapnia. The findings in an experimental model merit further animal investigations as well as a clinical reappraisal of the ventilatory management of sick newborn human infants.** ***Pediatrics* 1995; 95:868–874; *brain, carbon dioxide, hypoxia-ischemia, perinatal, rats.***

ABBREVIATIONS. CO_2, carbon dioxide; O_2, oxygen; CBF, cerebral blood flow; NMDA, N-methyl-D-aspartate; N_2, nitrogen.

Recent clinical investigations suggest that premature infants who require mechanical ventilation to prevent or minimize hypoxemia arising from respiratory distress syndrome are at increased risk for the development of either periventricular leukomalacia or periventricular/intraventricular hemorrhage if hypocapnia occurs during the course of respiratory management.[1–3] It is presumed that the hypocapnia occurs as a consequence of the hyperventilation necessary to maintain optimal paO_2. A causal relationship between low $paCO_2$ and hypoxic-ischemic or hemorrhagic brain damage was not established in any of the studies, but given the multiple effects of carbon dioxide (CO_2) on hemodynamics and metabolism (see Discussion), a cause and effect relationship is certainly plausible.

The clinical findings reported above are reminiscent of the alterations in systemic acid-base homeostasis that occur in immature rats undergoing hypoxic-ischemic brain damage.[4] When 7-day postnatal rats are exposed to 8% oxygen (O_2), the hypoxemia leads to a metabolic acidemia (lactacidemia).[5] The animals hyperventilate to an extent that produces hypocapnia; which, in turn, completely compensates for the metabolic acidemia; and blood pH remains normal. The question remains as to the contribution of the hypocapnia to the severity of hypoxic-ischemic brain damage. Accordingly, we devised a series of experiments whereby immature rats were subjected to cerebral hypoxia-ischemia with or without CO_2 added to the hypoxic gas mixture to which the animals are exposed.

MATERIALS AND METHODS

Dated, pregnant Sprague-Dawley rats were purchased from a commercial breeder (Charles River Laboratories, Wilmington, MA) and housed in individual cages. Offspring, delivered vaginally, were reared with their dams until time of initial experimentation at 7 days of postnatal age.

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Induction of Cerebral Hypoxia-Ischemia

To produce brain damage caused by cerebral hypoxia-ischemia, 7-day postnatal rats were anesthetized with halothane (4% induction; 1 to 1.5% maintenance) 30% O_2 balance nitrous oxide. Under anesthesia, the midline of the neck was incised in the longitudinal plane, and the right common carotid artery was identified and separated from contiguous sutures. The artery then was permanently ligated with 4-0 surgical silk and the wound sutured. Upon recovery from anesthesia, the animals were returned to their dams for 3 to 4 hours. Thereafter, they were placed in 500 mL air-tight jars partially submerged in a 37°C waterbath to maintain a constant thermal environment at 35°C ambient temperature. A humidified gas mixture of 8% O_2 balance nitrogen combined with a specific concentration of CO_2 (see below) was delivered into the jars via inlet and outlet portals at an approximate flow rate of 100 mL/min. The rat pups were exposed to the gas mixture for 2.5 hours, after which they were allowed to recover for 15 minutes in open jars in the waterbath. Thereafter, the animals were returned to their dams until 30 days of postnatal age at which time they were killed for neuropathologic assessment (see below). The combination of unilateral carotid artery occlusion and systemic hypoxia with 8% O_2 is known to produce brain damage in the form of selective neuronal necrosis or infarction predominantly of the cerebral hemisphere ipsilateral to the arterial ligation.[4,6] The brain of the 7-day postnatal rat is roughly comparable with that of a 32 to 34 week gestation human fetus or newborn infant; ie, cerebral cortical layering is complete, the germinal matrix is involuting, and white matter has yet to myelinate.

The O_2-CO_2-N_2 gas mixtures to which immature rats were exposed were calibrated commercially to ensure a precise inhalant concentration of oxygen. Some carotid artery ligated immature rat pups were exposed to a gas mixture containing 3.07% CO_2-7.83% O_2-balance N_2, while littermate controls simultaneously were exposed in separate jars to 7.94% O_2-balance N_2. Other rat pups were exposed to 6.19% CO_2-8.15% O_2-balance N_2 while littermate controls simultaneously were exposed to 7.90% O_2-balance N_2. The last group of rat pups was exposed to 8.98% CO_2-8.00% O_2-balance N_2, while littermate controls were exposed to 7.90% O_2-balance N_2. Accordingly, experimental and control littermates were exposed to O_2 concentrations varying by no more than 0.11%.

Neuropathologic Analysis

At 30 postnatal days (23 days of recovery), all animals were injected subcutaneously with a lethal dose of pentobarbital (100 mg/kg), and their brains were rapidly removed from the skulls and placed in solutions of formaldehyde, acetic acid, and methanol (FAM; 1:1:8). Two observers, blinded to the experimental manipulation, graded the presence and extent of brain damage in each animal, and individual brains were grouped to one of five categories as previously described:[7,8] Grade 0 = normal; Grade 1 = mild brain atrophy; Grade 2 = moderate brain atrophy; Grade 3 = atrophy with cystic cavitation <3 mm; and Grade 4 = atrophy with cystic cavitation >3 mm (Fig 1). Mild atrophy included those brains in which the posterior diameter of the ipsilateral cerebral hemisphere was up to 25% less than that of the contralateral hemisphere. Moderate atrophy included those brains in which the posterior diameter of the ipsilateral hemisphere was 25 to 50% less than that of the contralateral hemisphere but short of cystic infarction.

Morphometric evaluation of the severity of damage within each brain also was determined based on the degree of shrinkage of the affected cerebral hemisphere. Individual brains were transsected in the coronal plane at the mid-infundibular level, and the diameter (width) of each cerebral hemisphere determined, excluding the lateral ventricle, choroid plexus and any cystic area. The ratio of the width of the damaged (ipsilateral) cerebral hemisphere to that of the contralateral hemisphere then was calculated. We previously have demonstrated that the contralateral cerebral hemisphere can be used as an internal control, the size of which is comparable to that of an age-matched brain not subjected to unilateral cerebral hypoxia-ischemia.[9]

Randomly selected brains with mild-moderate atrophy were subjected to histologic analysis to ascertain whether the nature and distribution of brain damage incurred by hypoxia-ischemia alone differed from that which occurred with the additive effect of either normocapnia or hypercapnia. These brains were from animals that had received either 0, 3, 6, or 9% CO_2 (2 in each group).

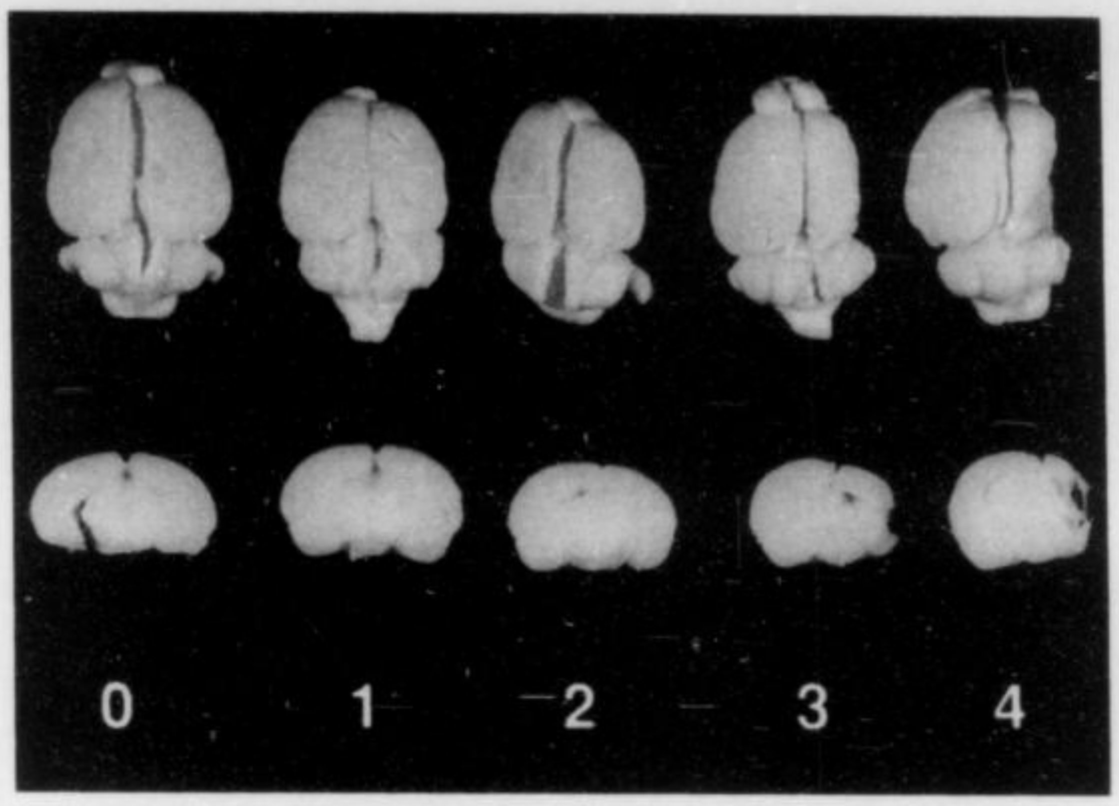

Fig 1. Representative brains from rats previously subjected to cerebral hypoxia-ischemia with or without supplemental CO_2 inhalation. Grade 0 = normal; Grade 1 = mild brain atrophy; Grade 2 = moderate brain atrophy; Grade 3 = atrophy with cystic cavitation <3 mm; Grade 4 = atrophy with cystic cavitation >3 mm.

Entire brains were transsected coronally at 2 mm intervals and embedded in paraffin. Histologic sections were prepared and stained with hematoxylin and eosin.

Blood Oxygen, Acid-Base, and Biochemical Analyses

In separate experiments, 7-day postnatal rats previously undergoing carotid artery ligation were exposed to humidified gas mixtures containing 8% O_2 and variable concentrations of CO_2, as described above. These animals were not placed in air-tight jars, but rather each rat pup was gently positioned head-first in the barrel of a 20 mL plastic syringe. The narrow anterior opening of the syringe was connected via polyethylene tubing to the gas tank, while the larger posterior opening remained unoccluded. A microthermister probe was positioned adjacent to the torso of each rat pup to maintain a constant ambient environment of 35°C by the use of a servo-controlled heating lamp positioned two feet directly above the syringe. Animals were exposed to hypoxia for either 1 or 2 hours, at which time their torsos were slowly withdrawn from the posterior end of the barrel to allow decapitation as the animal continued to inhale the gas mixture, an impractical procedure when the animals are exposed to hypoxia in jars (see above). Blood (approximately 0.1 mL) was immediately collected from the severed neck vessels for analysis of pO_2, pCO_2, and pH on a micro-blood gas analyzer (Model ABL-30; Radiometer America Inc, Westlake, OH). In separate animals, blood was collected and plasma separated for determination of glucose concentration on a micro-glucose analyzer (Glucostar; Beckman Instruments, Inc, Fullerton, CA). An additional aliquot of whole blood (0.02 mL) was diluted 1:10 in 0.5 M perchloric acid for later determination of blood lactate concentration.[10]

Statistical Analysis

The brain damage scores and the width ratio data were analyzed by analysis of variance models in which time (length of hypoxia-ischemia) and CO_2 concentration were fixed effects and litter was a random effect.[11] Because the data were unbalanced, the models were estimated by maximum likelihood and hypotheses were tested by likelihood ratio tests. Although the brain damage scores were nominally ordered categorical data, it was elected to treat them as continuous because: 1) the number of categories was relatively large (several animals were scored as intermediate between the predefined categories); and 2) litter effects are generally important in these studies; therefore, the available continuous data modeling methods were judged to be more reliable in handling such random effects. All analyses were executed in *S-Plus* version 3.1 (Statistical Sciences, Inc, Seattle, WA).

Institutional Approval

The experiments described here were reviewed by the Animal Care and Use Committee of The Milton S. Hershey Medical

Center—The Pennsylvania State University, and approved on October 1, 1992.

RESULTS

As previously mentioned, 7-day postnatal rats exposed to systemic hypoxia with 8% O_2 develop hypocapnia owing to hyperventilation.[5] Accordingly, our initial experiment was conducted on 39 similarly aged rat pups exposed to 8% O_2 combined with 3% CO_2, a concentration anticipated to maintain normocapnia; 37 littermate controls were exposed to 8% O_2 without added CO_2. One CO_2 exposed animal died during hypoxia-ischemia, while four hypoxic-ischemic controls died. Neuropathologic assessment at 30 days of age revealed a protective effect of CO_2, such that 30/38 (79%) rats showed either no or mild brain damage compared with 13/33 (39%) controls (Fig 2). Cystic infarction occurred in only four CO_2 exposed rat pups compared with 14 controls. Overall neuropathologic morbidity was significantly less in the CO_2 exposed animals (P = .001).

Given the protective influence of maintaining normocapnia in immature rats subjected to cerebral hypoxia-ischemia, we sought to ascertain whether or not hypercapnia would provide additional protection without increasing mortality. Two additional groups of 7-day postnatal rats were studied, one exposed to 6% and the other to 9% CO_2. Once again, littermate controls were exposed to systemic hypoxia alone. Mortality with 6% CO_2 was 1/21 compared with 6/23 controls, while mortality with 9% CO_2 was actually 0/23 compared with 4/19 controls. Neuropathologic assessment at 30 days indicated a dramatic protective effect of CO_2, such that at 6% CO_2 all 20 animals showed either no damage or mild atrophy ($P < .001$) and at 9% CO_2 19/23 (83%) animals showed no or mild damage ($P < .001$) (Fig 2).

A composite of the neuropathologic results is shown in Fig 3. The data show that the greatest reduction in hypoxic-ischemia brain damage occurred in immature rats exposed to 6% CO_2 with no apparent further protection at higher concentrations. To ascertain any difference in neuropathologic outcome produced by inhalation of high concentrations of CO_2, 24 7-day postnatal rats were subjected to cerebral hypoxia-ischemia, 12 of which inhaled 6% CO_2 and the remainder 9% CO_2-8% O_2-balance N_2 for 3 hours. All animals survived, and neuropathologic examination at 30 days of postnatal age revealed a mean damage score of 2.2 for the 6% CO_2 group, significantly less than the mean score of 3.7 for the 9% CO_2 group (P = .009) (Table 1). Accordingly, immature rats inhaling 9% CO_2 are less protected from hypoxic-ischemic brain damage than are rat pups inhaling 6% carbon dioxide.

Analysis of the diameters (widths) of the damaged (ipsilateral) cerebral hemispheres compared with respective contralateral hemispheres provided results near identical to those obtained upon visual inspection of the brains (Fig 4). Specifically, the ipsilateral/contralateral width ratio was lowest in rats previously exposed to 0% CO_2 and highest in those exposed to 6% CO_2. Ratios were comparable in the 3 and 9% CO_2 exposed animals.

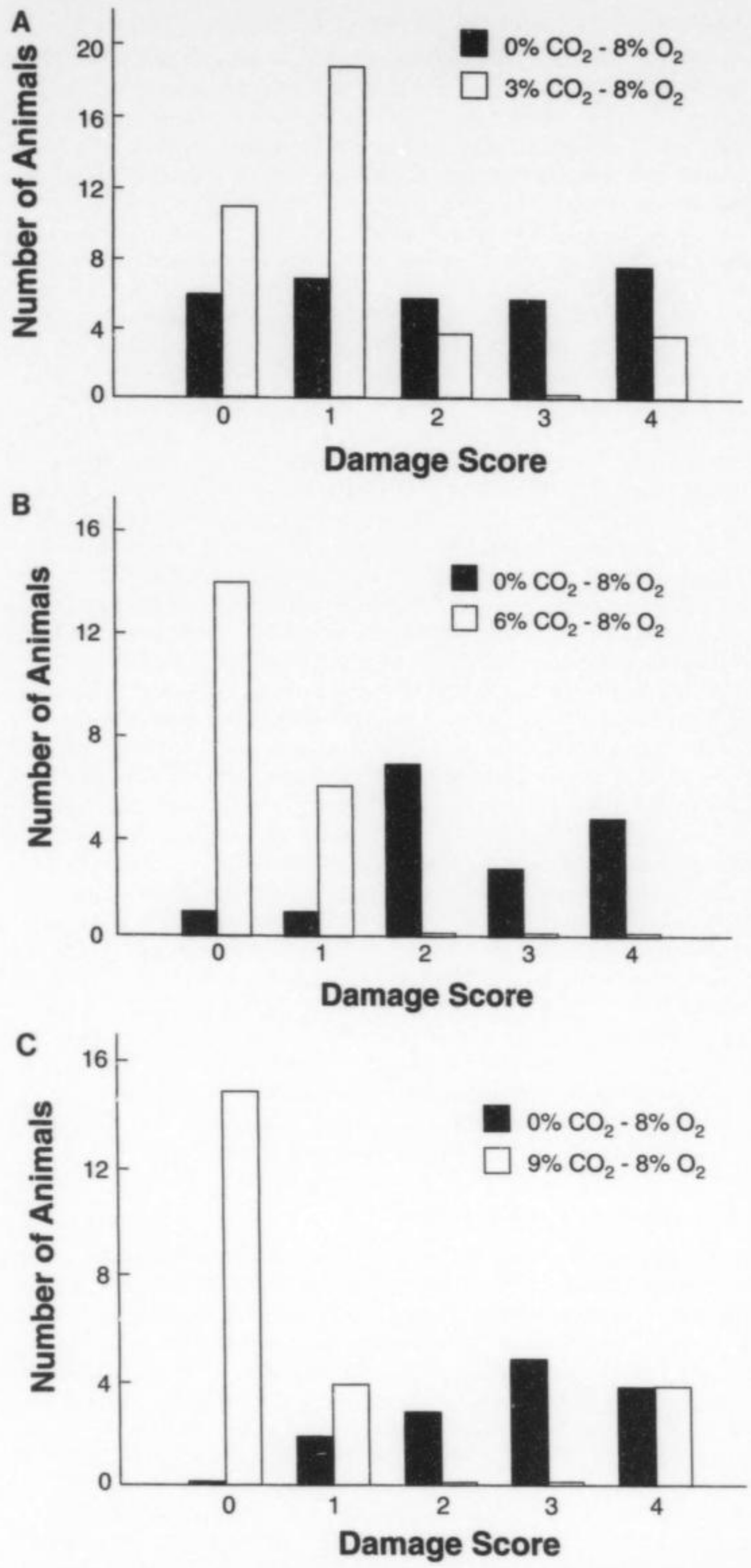

Fig 2. Severity of brain damage in rats previously subjected to cerebral hypoxia-ischemia with or without supplemental CO_2 inhalation. Seven-day postnatal rats underwent unilateral carotid artery ligation followed by inhalation of either 0, 3, 6, or 9% CO_2-8% O_2-balance nitrogen after which they were returned to their dams until 30 days of postnatal age. The extent of brain damage in each animal was determined as described in the text (see also Figure 1).

Histologic sections of selected brains from each CO_2 exposure group showed mild to moderate damage involving the cerebral cortex as combined laminar and columnar necrosis, hippocampus with relative sparing of the fascia dentata and medial CA3, dorsolateral striatum, and dorsolateral thalamus. There were no differences in the distribution of the damage or the quality of the cellular reaction among the various CO_2 groups examined.

Blood O_2 and acid-base status was determined in 7-day old carotid artery ligated rats subjected to

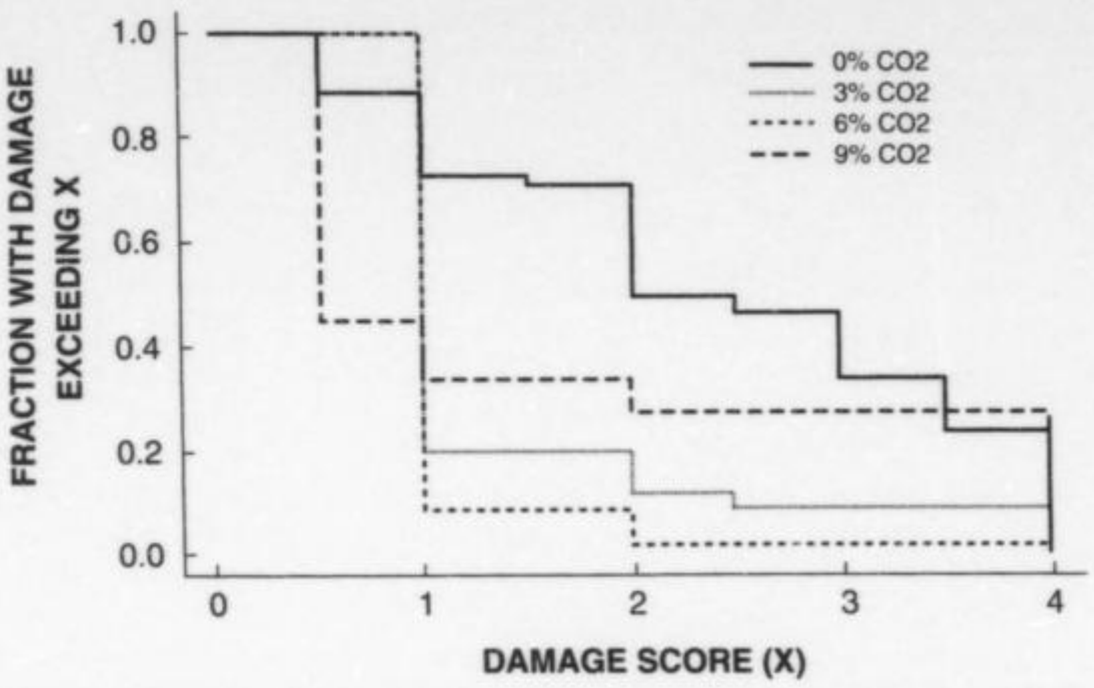

Fig 3. Empirical distribution function plot for neuropathology scores. This figure plots, for each possible damage score X, the fraction of rats in each CO_2 group for which damage score was greater than or equal to X. Thus, approximately 70% of animals in the 0% CO_2 group had scores of 2 or higher, 20% in the 3% group, 10% in the 6% group, and 35% in the 9% group.

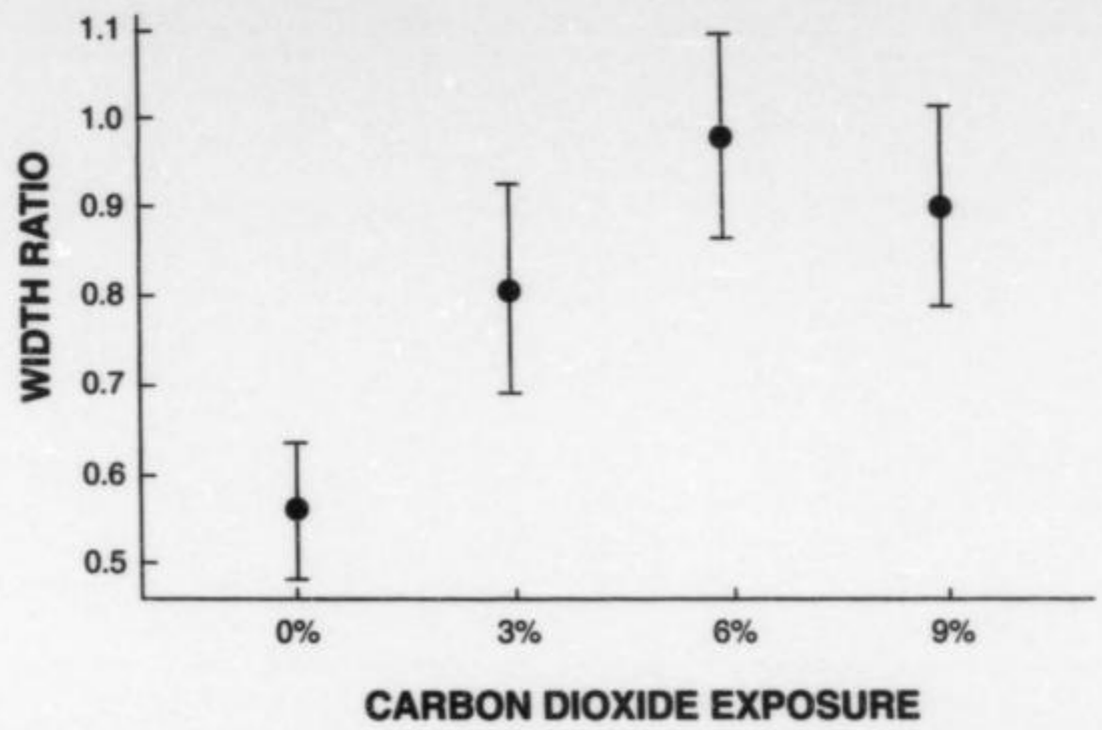

Fig 4. Ninety-five percent confidence intervals for mean interhemispheric diameter (width) ratios by CO_2 group. This figure displays 95% confidence intervals for the mean ratio of the width of the ipsilateral cerebral hemisphere/width of the contralateral hemisphere, estimated under the linear model with random litter effects (see Statistical Analyses). The ● indicates the estimated mean, and the vertical line denotes ±2 SEM.

TABLE 1. Severity of Brain Damage in Rats Previously Subjected to Cerebral Hypoxia-Ischemia With Either 6% or 9% CO_2 Supplementation

Grade of Brain Damage	6% CO_2	9% CO_2
1	4	0
2	5	4
3	2	2
4	0	0
5	1	6

Values represent numbers of animals in each category. Grades of brain damage are described in the text.

systemic hypoxia with or without carbon dioxide inhalation (Table 2). Hypoxia without supplemental CO_2 was associated with hypocapnia to the extent that the concurrent metabolic acidemia was completely compensated, resulting in a normal blood pH. Hypoxia with 3% CO_2 maintained normocapnia, but once again pH was unchanged from control, owing in this instance to the absence of a metabolic acidemia. Higher concentrations of CO_2 caused progressively greater hypercapnia; which, despite the development of a metabolic alkalosis, was associated with a reduction in blood pH.

CO_2 exposure had a dramatic effect on blood lactate, such that progressively higher pCO_2 (Table 2) was associated with progressively lower concentrations of the metabolite, even to levels below control (Fig 5). CO_2 inhalation also influenced plasma glucose concentrations, preventing entirely the typical mild hypoglycemia seen at 2 hours of hypoxia-ischemia and actually causing hyperglycemia, especially in those animals exposed to higher concentrations of the gas.

DISCUSSION

The findings of the present investigation are straightforward to the extent that normocapnic cerebral hypoxia-ischemia is associated with less severe brain damage than seen in hypocapnic cerebral hypoxia-ischemia. Furthermore, at least mild hypercapnia is more protective to the immature brain subjected to hypoxia-ischemia than normocapnia. The protective influence of CO_2 inhalation was not associated with an increased hypoxic-ischemic mortality, as animal survival was unaffected even at blood pCO_2 values as high as 75 mm Hg.

Numerous mechanisms potentially exist whereby CO_2 protects the immature brain from hypoxic-ischemic damage. Such mechanisms would entail hematologic, cardiovascular, cerebrovascular, and metabolic factors. An increase in blood pCO_2 (or a decrease in pH) shifts the oxygen-hemoglobin dissociation curve to the right (Bohr effect), the result of which is a decrease in the affinity of hemoglobin for oxygen (for review, see Reference 12). Therefore, at the capillary level (brain), CO_2 would tend to raise pO_2, increase the gradient for any given oxyhemoglobin saturation, and facilitate transfer of O_2 to the tissue for oxidative processes. Whether or not the phenomenon of increased capillary pO_2 at a constant peripheral pO_2 occurred in immature rats breathing 8% O_2 cannot be determined, owing to the confounding variable of progressively decreasing metabolic acidemia with increasing CO_2 exposures (see Table 2). Accordingly, blood pH remained relatively constant despite increasing pCO_2 to tensions as high as 61 mm Hg (6% CO_2 exposure), tensions which were highly protective to immature rat brain undergoing hypoxia-ischemia.

CO_2 might also have preserved cardiac function during systemic hypoxia. In our immature rat model of hypoxic-ischemic brain damage, systemic hypotension occurs as a presumed consequence of hypoxic cardiac depression;[5] the hypotension, in turn, is associated with reduced blood flows to the vulnerable structures of the brain.[13,14] Experiments from our laboratory in paralyzed and artificially ventilated newborn dogs subjected to graded systemic hypoxia (5 to 8% O_2) indicate that progressive systemic hypotension and ultimately cardiac arrest occur in direct relationship to worsening metabolic acidemia (lactacidemia).[15] In the present investigation of immature rats, lactacidemia was reduced or minimized by CO_2 inhalation. The inhibition of systemic lactate

TABLE 2. Blood Oxygen and Acid-Base Status of Immature Rats Exposed to Hypoxia and Carbon Dioxide

Variable	pH	pCO_2 (mm Hg)	pO_2 (mm Hg)	HCO_3^- (mmol/L)	Base Excess
Control	7.40 ± 0.01	44.2 ± 1.7	60.2 ± 3.8	27.3 ± 0.9	2.6 ± 0.8
Hypoxia (1 h)					
0% CO_2	7.42 ± 0.01	26.1 ± 1.0†	32.8 ± 2.2†	17.5 ± 0.6†	−5.8 ± 0.7†
3% CO_2	7.40 ± 0.05	38.6 ± 1.3§	32.3 ± 1.1†	25.7 ± 1.5[a]	2.5 ± 1.6[a]
6% CO_2	7.34 ± 0.02†[a]	54.8 ± 2.0*§	35.7 ± 1.9†	28.3 ± 2.0*§	2.6 ± 1.5[a]
9% CO_2	7.27 ± 0.01†§	69.0 ± 1.9†§	38.8 ± 1.0†	30.0 ± 0.9†§	3.0 ± 0.8§
Hypoxia (2 h)					
0% CO_2	7.42 ± 0.03	25.4 ± 1.2†	35.6 ± 3.0†	16.4 ± 1.0†	−7.1 ± 0.9†
3% CO_2	7.42 ± 0.01	39.5 ± 1.2§	31.6 ± 2.0†	26.1 ± 0.8§	1.6 ± 0.8§
6% CO_2	7.33 ± 0.01†[a]	57.2 ± 0.8†§	36.1 ± 2.1†	29.4 ± 0.9§	3.4 ± 0.9§
9% CO_2	7.26 ± 0.01†§	67.5 ± 1.8†§	34.9 ± 0.8†	29.4 ± 1.2§	3.1 ± 1.1*§

Values represent means ±1 SEM for five animals in each group. Control rats (n = 11) underwent neither carotid artery ligation nor systemic hypoxia.

* $P < .05$, † $P < .001$ compared with control.

[a] $P < .05$, § $P < .001$ compared with 0% CO_2 at same interval of hypoxia.

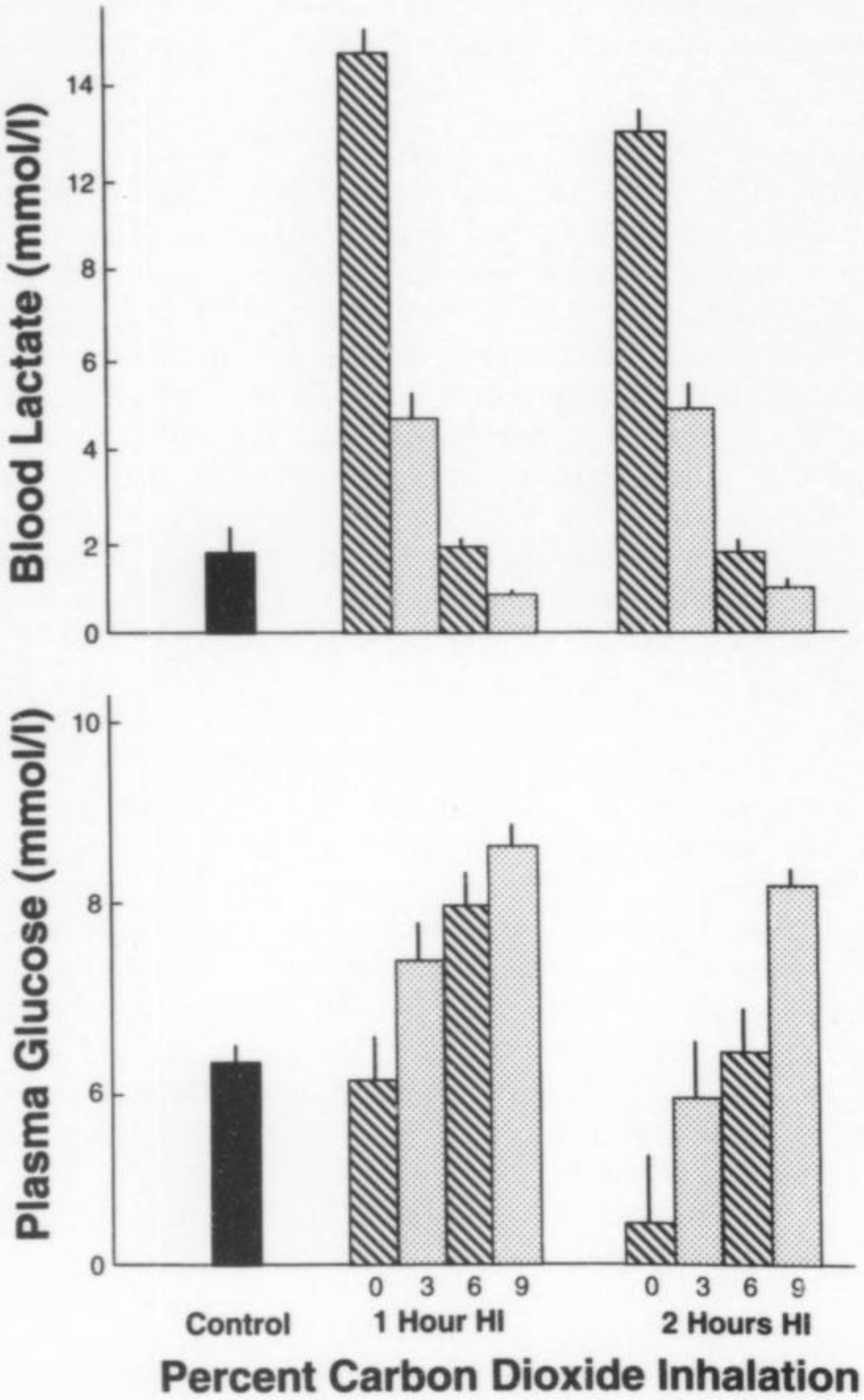

Fig 5. Plasma glucose and blood lactate concentrations in immature rats exposed to hypoxia with or without supplemental CO_2 inhalation. Bars represent mean concentrations of 5 to 6 animals; vertical lines denote ±1 SEM.

production by CO_2 inhalation during hypoxia would serve to maintain optimal cardiovascular function, thereby preventing the occurrence or reducing the severity of cerebral ischemia necessary for brain damage to occur.

Cerebral vascular reactivity is strongly influenced by CO_2. Hypocapnia lowers cerebral blood flow (CBF), while hypercapnia increases CBF (CO_2 – CBF sensitivity), and these responses to CO_2 are active in immature animals, albeit less well developed than in adults (for review, see Reference 15). Accordingly, it is possible that the anticipated hypoxic dilation of the cerebral vascular bed is attenuated to some extent by the concurrent hypocapnia that occurs in immature rats exposed to 8% O_2 without supplemental CO_2.[5] In this regard, Ruta et al[16] recently showed in adult rats that middle cerebral artery occlusion combined with hypocapnia is associated with a greater reduction in blood flow to a wider area of the ipsilateral cerebral hemisphere than occurs with middle cerebral artery occlusion alone, suggesting that hypocapnia increases the size of the brain region at risk for infarction. Accordingly, increasing pCO_2 might serve to maximally dilate the cerebral vasculature in the setting of hypoxia, thereby improving O_2 delivery to the cerebral hemisphere even when its major nutrient artery (common carotid) has been previously ligated.

Finally, CO_2 has several actions on cerebral metabolism *per se*. Hypercapnia leads to a generalized suppression of oxidative metabolism, as indicated by an inhibition of cerebral energy utilization and a curtailment of glycolytic flux.[17–20] Tissue lactate production also is reduced, at least under physiologic conditions.[21–23] During anoxia in newborn rats, hypercapnia is associated with a greater preservation of high-energy phosphate reserves and energy expenditure than during normocapnic anoxia.[24] The combined effect of enhanced substrate (glucose) delivery (see above) and reduced energy demand would serve to maximally protect the vulnerable tissue from energy failure and ultimate damage during hypoxia-ischemia.

Lastly, recent in vitro investigations from several laboratories have shown that acidosis of the brain extracellular compartment blunts the neurotoxicity of excitatory amino acid neurotransmittters, especially glutamate, via an inhibition of the N-methyl-D-aspartate (NMDA) cell surface receptor.[25–27] Furthermore, hypercapnia actually reduces brain tissue concentrations of glutamate[22,23,28] and possibly inhibits the secretion of the excitatory amino acid into the synaptic cleft during hypoxia-ischemia. The manner in which extracellular acidosis inhibits NMDA receptor activation during hypoxia-ischemia is not entirely clear, but in vitro studies suggest a down regulation of NMDA-gated ion channel currents with a secondary reduction in intracellular calcium accumulation (for review, see Reference 29).

We do not have a good reason why the immature rats exposed to 9% CO_2 fared less well than the rat pups breathing 6% CO_2, as the extent of neuroprotection of the former group was comparable with that of the normocapnic animals and better than that of the hypocapnia animals. Rat pups exposed to 9% CO_2 exhibited the lowest blood pH of any group, owing exclusively to hypercapnia, since calculated HCO_3^- and measured blood lactate concentrations were greater than and less than control, respectively (see Table 2 and Figure 5). Possibly, the respiratory acidosis combined with hypoxia leads to a depression of cardiovascular function similar to that of the normocapnic, hypoxic rat pups; leading, in turn, to comparable degrees of cerebral ischemia involving the cerebral hemisphere ipsilateral to the carotid artery occlusion. Measurements of regional CBF during the course of hypoxia ischemia will be required to resolve the issue.

The present findings in immature rats to some extent can be extrapolated to the human situation, especially as regards premature infants undergoing mechanical ventilation. However, there are necessary precautions. The brains of rats at 7 days of postnatal age are histologically comparable with those of 32 to 34 week gestation newborn infants, but a species difference exists in their brains' vulnerability to hypoxia-ischemia. When brain damage occurs in these age infants, the injury is focused predominantly on subcortical and periventricular white matter, occasionally with involvement of the overlying cerebral cortex. Less mature infants exhibit damage largely restricted to the periventricular regions (for review, see Reference 30). In immature rats, white matter necrosis occurs largely in association with gray matter infarction.[4,6] In the present investigation, the rats were not mechanically ventilated but rather spontaneously hyperventilated in response to systemic hypoxia. In premature infants, hypocapnia occurs as a consequence of mechanical ventilation, which might represent a confounding variable in the pathogenesis of hypoxic-ischemic brain damage. However, Graziani et al[2] found a correlation between hypocapnia and the ultimate occurrence of cystic periventricular leukomalacia and cerebral palsy in their analysis of 192 premature infants, all of whom were mechanically ventilated.

Obviously, further investigations are required to confirm the present findings in other animal models and to elucidate those physiologic and biochemical mechanisms whereby CO_2 protects the perinatal brain from hypoxic-ischemic damage. Whatever the outcome of such experiments, the observation remains that hypocapnia aggravates perinatal hypoxic-ischemic brain damage while mild hypercapnia is neuroprotective, at least in the immature rat. Given the frequent occurrence of hypocapnia during mechanical ventilation of premature infants suffering respiratory distress syndrome or other pulmonary disease and the apparent association between hypocapnic hypoxia-ischemia and ultimate cerebral palsy, the present findings in an experimental model merit corroboration in other animal models as well as a clinical reappraisal of the ventilatory management strategies of sick newborn human infants.

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EFFECT OF REACTIVE OXYGEN SPECIES ON THE ERYTHROCYTE CALCIUM-PUMP FUNCTION IN PROTEIN-ENERGY MALNUTRITION

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Abstract. Marasmus is felt to result from a diet that is fairly equally deficient in protein and in calories. In contrast, kwashiorkor is usually explained as being caused by a diet deficient more in proteins than in calories. This explanation for kwashiorkor has been questioned by Golden and Ramdath,[1] who suggested that the pathogenesis could be better explained by an imbalance between the quantity of free oxygen radicals and scavengers of those free oxygen radicals. The authors of this paper took anemia as one of the usual clinical features of kwashiorkor and attempted to verify this hypothesis by measuring "the presence of non-heme iron and endogenous products of lipid peroxidation in the erythrocyte membranes of normal and kwashiorkor subjects and assessed the susceptibility of the membranes to exogenously generated reactive oxygen species." The extent of lipid peroxidation in the erythrocytes of kwashiorkor subjects was three times that found in erythrocytes of normal subjects. The erythrocytes from kwashiorkor subjects were much more susceptible to oxidative stress than were the erythrocytes from normal subjects and the oxygen free radicals reduced the erythrocyte calcium-pump function in kwashiorkor subjects to a greater extent than in normal subjects. These findings lend support for the hypothesis of the pathogenesis of kwashiorkor proposed by Golden and Ramdath.

Commentary: If evidence continues that the imbalance between free oxygen radicals and scavengers is responsible for kwashiorkor, the treatment for this type of malnutrition will need to be modified. Treatment in the future may include the elimination of intestinal organisms that stimulate the production of free oxygen radicals. Iron therapy may be detrimental, as iron stimulates the redox cycle and, in turn, the production of free oxygen radicals. Treatment may include the use of free oxygen radical scavengers such as Vitamin A or E or beta carotene. This is an important paper and we are sure to see more information on the hypothesis. *Biosci Rep*. 1992;12:433–443.

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